AGI ALPHA: Verified Repair and Acquired Transfer Under Strong Predictive Controls

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Abstract

AGI ALPHA aims to construct verified instruments that improve subsequent invention and support useful, economically viable delivery. This edition investigates that aim using two real perturbation-data families and a bounded executable representation language. It continues a verified release whose inherited memory matched ordinary reuse and conventional construction, rather than improving on them. The new work qualifies actual data interfaces, preserves an ordinary-extraction failure, generates arithmetic representations from observed single-intervention effects, and repairs a consequential exchange-symmetry defect through an independently reproduced constraint. These are distinct engineering and experimental contributions. On 44 protected Norman pairs and 58 protected Wessels pairs, the repaired constructor fails the declared advantage criterion against strong conventional controls. The fresh inherited construction beats additivity but loses to matching mean and full-information ridge; equivalent ordinary reuse gives identical predictions and generated policy selection retains the incumbent. The resulting engineering qualification does not establish a previously inaccessible biological capability, improved invention policy or a commercial flywheel.

1. Objective, inherited evidence and authority

The intended progression is an enabling construction, accepted useful delivery, complete-cost surplus, unrestricted earned resources, further capability construction and a fresh useful successor. The central empirical question is whether the connected mechanisms improve construction against complete competent alternatives after acquisition and verification costs. Publishing software or an article does not complete that progression.

The continuation baseline is Lab PR #8, merged at bef0b697206699330a158f1430fe3b3ca58c6d97. Its published source-and-evidence ZIP is 5,858,230 bytes with SHA-256 9e80e3708f430b0a5ac18c8580eb2320fe bb213cbda63cc7eaa00be81b78e95d. The baseline's published owner download receipt records a validated extraction matching 5,289 payload identities and 50 frozen sources, 94 unit tests, 11 verifier tests and the original/repaired verifier-defect regression. This is the owner's 16-gate execution, not the same independent execution identified in the continuation brief. In this cycle, the separate verifier again checked all original payload identities, ran the full unit suite on ordinary extraction, and rechecked three archive tests on validated extraction. That narrower latter branch is not a new independent full-94 qualification. The platform reported `immutable: false`; this is a hash-identified prerelease, not platform-enforced immutability. Historical releases and adverse results are retained.[1]

The inherited eight-case comparison accepted 8/8 with structured inheritance, 8/8 with equivalent-content ordinary reuse and 8/8 with conventional construction; reset and operator removal accepted 0/8. Those cases belong to one bounded local-file protocol family. The contrast establishes the contribution of acquired content relative to its removal, while the ordinary-reuse tie prevents a structured-memory superiority claim. The original policy improvement was developer-written. None of these results established general invention, complete-cost advantage or earned funding.

[1] MONTREAL.AI, mechanism-completion release, merged implementation, and separate download audit, 29 October 2026.

Cycle 006 declared 120 wall minutes, including a 25-minute verification/publication reserve; 5,400 experiment CPU-seconds; 6 GiB of new artifacts; a 32 GiB received-network ceiling; 4 GiB per heavy worker; at most two concurrent heavy workers; and four substantive repair rounds. Paid model calls, spending, customer outreach, actual transfers and account/security changes were not authorized. All commercial transactions in this edition are simulated. Unavailable agent-inference billing remains unknown.

2. Selection and actual data qualification

The screen compared three materially different possibilities before selecting a bounded test. It did not admit a commercial flagship merely because data were convenient.

Possibility	Missing-means hypothesis	Complete alternative and decision
Combinatorial cellular response	Construct representations that predict useful non-additive responses from limited prior interventions.	Existing additive, linear, GEARS, Systema and calibrated perturbation workflows require strong controls. Execute a separately registered real-data pilot; novelty remains unestablished.
Cross-model metabolic repair	Transfer a local direction repair to remove whole-model energy-generating behavior.	Existing GlobalFit-style whole-model methods already address the proposed restriction; retained counterexamples prevent a local success being promoted to a global cure.
Constitutive identification	Recover absolute mechanical properties without an apparently required force measurement.	Missing physical information can make inversion nonunique, while established load-plus-displacement methods supply an alternative. No qualified physical measurement is supplied here.

2.1 PSGRN was checked, not assumed installed

At PSGRN revision 74aa640f7c472b23a69811f6795bb17678efd344, the documented custom-model file is `src/main.py`; the previously copied `src/main_fast.py` does not exist. An actual application-entrypoint probe reached `causalscbench/apps/main_app.py` and refused at missing `slingpy`. The isolated inspection environment is not the original complete PSGRN environment. The full model and benchmark were not executed.[2]

Read-only HDF5 range access qualified the mirror's metadata while preserving TLS verification and rejecting wrong or unbounded responses. The 8,700,873,216-byte mirror exposes a contiguous float32 matrix of 247,914 cells by 8,749 genes. Eight ranges retrieved 8,178,176 bytes. Metadata contains 11,485 controls and 651 genes eligible under the pinned greater-than-100-intervention-cell condition before the unexecuted strong-effect filter. This implies 423,150 directed pairs, rather than counting all raw columns as eligible perturbation genes.

The full mirror SHA-256 and equivalence to the original Figshare input remain unknown. More importantly, the upstream normalization depends on full-matrix row totals and their global median. Arbitrary cell/gene truncation or a stored UMI count is not a certified replacement. Removing local whole-file storage for metadata inspection is useful ordinary library composition; it is not a new biological means of inference. These findings motivated a distinct experiment rather than a retrospective success claim on PSGRN.

[2] GuanLab, PSGRN README at the qualified revision.

2.2 Source-bound Norman panel

The selected pilot uses the authors' published `source_data/suppl-qqplot.xlsx` at revision bfa6eeea2bd145a1af2ec0127a2e808cc38456a9. Its 10,905,627 bytes match SHA-256 1b652c352d4dabdd20324d7fd74d331898fa22c51cf6fa28ff65aa1322a6c78f. The export contains 124 double-intervention conditions and 1,000 readout genes, with observed A, B, AB and control means. Its readouts were selected by control-expression rank, not by the magnitude of the observed combination effect. The source code and notices establish how this narrower panel was constructed.[3]

The file supplies condition means rather than raw cells, biological replicates or variance estimates. The 124,000 rows are not independent experimental tasks. Shared single-intervention and control measurements, correlated genes and published preprocessing remain dependencies. This pilot does not reconstruct the original paper's training assignments or establish exact equivalence to a differently processed Norman input.

The verifier selected 80 development pairs and 44 protected pairs. The development lane contains 60 fit and 20 calibration pairs. Protected double-intervention outcomes are withheld from current construction. As the source is public, prior model exposure cannot be excluded; this is role-blind current development, not a claim that no model has previously encountered the observations.

2.3 Current alternatives constrain the claim

Simple baselines are substantive competitors, not ceremonial controls. The 2025 linear-baseline comparison, Systema's perturbation-specific evaluation, GEARS' multigene predictions, and a later 27-method/29-dataset benchmark all predate this cycle.[4][5][6][7] A current calibration study further cautions against using the near-additive Norman benchmark alone and provides positive-control methodology.[8] This literature does not establish the result of our candidate, but it prevents presenting residual prediction, metric calibration or multigene inference as an undiscovered class of work. Methods not actually run remain unqualified controls rather than silently weak substitutes.

2.4 A distinct successor-data family

The complete 219,393,529-byte `WesselsSatija2023.h5ad` from the versioned scPerturb record matched its published MD5 and independently computed SHA-256. The data owner inspected metadata without consuming combination outcomes. The raw file contains 30,707 cells, 21,052 genes, 28 single interventions, 158 canonical distinct intervention pairs and 424 control cells. Its metadata identifies THP-1 monocytes and CRISPR-cas13. The inspected benchmark YAML's HEK293T description and the paper's processed 157-pair cohort are not silently inherited.[9]

This creates a separately identified raw-data pilot opportunity, not reproduction of the publication's processed benchmark. The verifier selected 100 development pairs (75 fit, 25 calibration) and 58 protected pairs before reading combination counts. It normalized each cell over all 21,052 genes to 10,000 counts, applied log1p, and selected the 1,000 highest-mean readouts using only the 424 control cells. The exported axis is canonical

[3] C. Ahlmann-Eltze, W. Huber and S. Anders, *Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines*, Nature Methods 22, 1657-1661 (2025), doi:10.1038/s41592-025-02772-6. Archived source and outputs; exact inspected repository revision.

[4] C. Ahlmann-Eltze, W. Huber and S. Anders, *Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines*, Nature Methods 22, 1657-1661 (2025), doi:10.1038/s41592-025-02772-6. Archived source and outputs; exact inspected repository revision.

[5] R. Vinas Torne et al., *Systema: a framework for evaluating genetic perturbation response prediction beyond systematic variation*, Nature Biotechnology, doi:10.1038/s41587-025-02777-8, published online 25 August 2025.

[6] Y. Roohani, K. Huang and J. Leskovec, *Predicting transcriptional outcomes of novel multigene perturbations with GEARS*, Nature Biotechnology 42, 927-935 (2024), doi:10.1038/s41587-023-01905-6, published online 17 August 2023.

[7] *Benchmarking algorithms for generalizable single-cell perturbation response prediction*, Nature Methods, doi:10.1038/s41592-025-02980-0, published 11 December 2025.

[8] H. E. Miller et al., *Deep learning perturbation models can outperform baselines on calibrated metrics*, Nature Biotechnology, doi:10.1038/s41587-026-03307-w, published 1 October 2026. Its complete-method and dataset-specific results are prior art, not experimental results of this laboratory.

[9] scPerturb, versioned source record 13350497, `WesselsSatija2023.h5ad`, CC-BY 4.0; S. Peidli et al., *scPerturb: harmonized single-cell perturbation data*, Nature Methods, doi:10.1038/s41592-023-02144-y. The source contribution and all preprocessing changes must be attributed with derived numeric artifacts.

rather than rank-ordered; zero-library cells would remain zero and none occurred. Only development outcomes and protected single-input features reached construction. Prior art reports useful nonlinear predictions on a processed Wessels cohort, while PRESAGE code/weights were not executed here; its noncommercial model license is separate from the public data license.

A remote processed-file HDF5 callback probe exited 139 without a result; its cause remains unresolved and the failed transfer is retained. The ordinary verified bulk download and native HDF5 driver provide the available alternative. That result argues for dependency removal, not a universal prohibition on range access. The raw-data download took 87.597 seconds wall time and 2.971 waited-child CPU-seconds. Input identity and metadata access alone establish no prediction, inheritance or economic advantage.

3. Executing construction and verification

3.1 Generated representations and bounded execution

The domain constructor begins with observed single-intervention effects relative to the same control. It fits centering, scale and three principal coordinates using development inputs only, with deterministic component-sign conventions. Arithmetic trees combine the resulting six A/B coordinates. The finite grammar supplies addition, subtraction, multiplication, bounded division, minimum and maximum; trees have bounded depth and node counts. At most eight generated basis functions augment a residual ridge model over the additive prediction. PCA, ridge, normalization and arithmetic primitives are developer-authored. Runtime-generated compositions are not unrestricted invention.

`mechanisms/policy006.py` generates versioned arithmetic policies through bounded synthesis/mutation. Its generic action-selector interface proposes measurement, representation, construction or repair action identities under explicit budgets; a trusted broker owns actual effects. The Norman constructor uses generated trees as representations through an explicit developer-supplied terminal correspondence. That use is distinct from proving that a learned general discovery policy has improved.

The historical Norman proposal trace used a hash-qualified host copy of Omega's original `Program.create` and `mutate`. Its recovered archive has no established module license, so that module is not vendored or represented as installed by the public reproduction. The public historical run instead replays the exact recorded native proposals. A separately authored bounded fallback generator executes as a separately labeled development experiment; it is not native-regeneration parity or live-model performance. Source/archive identities and this licensing boundary remain part of provenance.

Generated arithmetic representations execute as JSON data in the qualified seccomp interpreter. They cannot import arbitrary generated Python or choose host files, credentials or network endpoints. Trusted array fitting and data acquisition occur outside that interpreter and are identified as trusted application effects. The trusted supervisor records resource receipts and applies per-process limits. Subsequent review found standalone termination and fast-exit storage-enforcement defects, documented in Section 5. This supervisor is separate from the generated-program seccomp boundary and does not establish exact escaped-descendant CPU accounting.

3.2 An observed criticism becomes a general construction

The first nominated representation depended on the arbitrary order of the two simultaneous intervention labels. Swapping A and B changed a prediction by as much as 0.430802 on exposed development cases. The initial grammar permitted ordered representations and did not impose the unordered-input contract. A separate verifier reproduced the actual prediction defect and emitted an established exchange-symmetry constraint bound to source, artifact and data identities.

The structural donor is the established finite-group averaging construction. The supplied source graph represents two exchangeable inputs contributing to a joint output. The matcher returns both permitted input correspondences, rather than manually selecting a desired A/B labeling. Simultaneous unordered inputs and matching units are applicability conditions. Ordered interventions do not admit this action.

The constructor consumes the adjudicated constraint by replacing each generated basis with

$$\phi_{\mathrm{sym}}(A, B) = \tfrac{1}{2}\{\phi(A, B) + \phi(B, A)\}.$$

It then refits using the same development data and unchanged prediction objective. This is a generic transformation of every generated basis, not a patch for one named pair. Group averaging is known mathematics and its implementation was a supplied primitive; the executed contribution is the constraint-triggered application to the selected generated artifact, not autonomous invention of the primitive.

Development observation	Original	Constructed repair
Maximum absolute swap difference	0.430802	0
Median calibration-pair RMSE	0.078464	0.079215
Pooled calibration RMSE	0.090095	0.089612

Restoring the invariant did not uniformly improve accuracy: the median worsened while pooled error decreased. This adverse contrast is retained. The mapping-consuming repair run took 1.885 seconds wall time and 1.875 waited-child CPU-seconds; these figures are not complete acquisition or monetary costs.

3.3 Retention acquires a generative operator

The new capability record binds an executable instrument, an applicability model and a generative procedure. Acquisition checks the actual selected pre-repair and post-repair configurations and the adjudication/verification identities. It retains generated basis structure plus the established symmetry constraint, rather than copying a success flag or retaining fitted output coefficients. The successor procedure generates related configurations and refits on target development data. The executed acquisition independently recomputed the original and repaired swap predictions, then stored and retrieved the record through the existing immutable memory implementation. Source corruption, stale applicability, poisoned arithmetic content and ordered/incompatible targets refuse. The acquired record contains no fitted PCA or output weights; its content hash is 7ea0e3fbd36db5480545b717fb1d94dace92a315a4f3998dbd6f0689f6f4dc35.

Equivalent-content ordinary reuse receives the same capability content and checks. Removing the operator leaves stopping/additive construction eligible. A new starting tool, a better policy and an advantage of structured retrieval are separate hypotheses.

3.4 Actual fresh construction and policy/toolbox cross

The Wessels successor executes the memory retrieval and related-generation path, fits a new latent representation and coefficients to the new gene axis, and produces predictions for 58 feature-only protected pairs. Eight generated priority programs compete through replay of exposed Norman development history. The selected priority is exactly the existing gain/cost rule. This is a null result for improved discovery policy, not a new policy win.

The four old/new policy by old/new toolbox cells, equivalent-content ordinary reuse, operator removal, frozen initial state, reset-and-refit state and history-best conventional action ordering receive up to four construction fits each, followed by one charged all-development refit. Every fit and rejected representation is retained. Old-tool recipes sampled by the policies violate exchange symmetry and fall back to explicit conventional additive stopping. Frozen/reset arms restore the initial construction catalog while retaining the same exposed history; they are toolbox ablations rather than complete independent lifecycle rebuilds.

On exposed Wessels calibration, the two policy arms with acquired tools, equivalent ordinary reuse and history-best ordering tie at median pair RMSE 0.119045; operator removal/additive stopping yields 0.133949. The selected recipe is the inherited repaired basis, refitted to fresh data, rather than a winning new basis mutation. The full cross took 44.308 seconds wall time and 44.270 waited-child CPU-seconds. This development contrast alone cannot distinguish a useful learned representation from its ordinary per-gene residual intercept. The independent conventional residual-mean and residual-ridge controls are consequently decisive.

4. Prospective protected specification

The registered primary endpoint is RMSE across the unchanged 1,000-gene panel for each protected pair. Acceptance requires at least 5% median pair-relative RMSE improvement against every qualified named control, improved pooled RMSE against every such control, at most 10% ninetieth-percentile pair-relative loss increase, and finite outputs with exact pair/gene identities. The complete controls are exact A+B-control, Systema MatchingMean, exactly (A+B)/2 when both singles are observed; full multioutput linear-kernel ridge on all 1,000 symmetric single-effect features; and additivity plus the per-gene development-mean residual. The fourth control was added before protected execution because the constructed model's unpenalized intercept could otherwise explain apparent benefit. The full ridge control standardizes the complete symmetric vector A+B-2control using fit data only. Both direct and additive-residual targets search nine ridge strengths from 1e-4 through 1e4 on the declared fit/calibration split, with deterministic tie rules, before all-development refitting. Direct ridge, residual ridge and training-mean controls are also reported separately. Hyperparameters are selected before protected outcomes, using 60/20 Norman and 75/25 Wessels fit/calibration pairs. A coordinatewise ridge does not qualify this control.

The secondary endpoint ranks observed absolute deviations from additivity, using a fixed top-100 relevance set, NDCG@50 and source/pair/gene hash tie-breaking. These are observed condition-mean deviations. They are not biological interaction significance, a cellular false-positive rate or a clinical endpoint.

Candidate, comparators, verifier, protocol, task identities and budgets are frozen before the fixed protected panels are evaluated. No scientific confirmation alpha is newly spent. The practical thresholds describe this finite cohort; correlated means and shared measurements do not support gene-independent standard errors or biological-population confirmation. Outcomes used for repair become exposed and cannot retain confirmation status.

4.1 Frozen outcomes: conventional controls remain stronger

Before either protected outcome panel was opened, the candidate/control freeze was published at revision 2802fd573bd8ae8cc5585f7e98108660927fb0da, with SHA-256 0f646f3481dd32e5a297115ea49d58615de48a4762 6390293bd167fd50ff7847. The independent checker re-executed the direct controls and complete development ridge selection, reproduced model predictions and verified exchange invariance before reading truth. The fixed runs completed once. All 102 pair outcomes, adverse cases and source-bound predictions are retained and now exposed for reproduction; they are unavailable as fresh confirmation for later repairs.

Frozen arm	Norman median / pooled RMSE	Wessels median / pooled RMSE
Constructed candidate	0.071083 / 0.105698	0.121445 / 0.132249
Exact additive	0.072337 / 0.098468	0.136662 / 0.149529
MatchingMean	0.083743 / 0.101691	0.105947 / 0.117994
Development-selected full ridge	0.073758 / 0.079619	0.065302 / 0.081320
Additive + mean residual	0.072179 / 0.095742	0.124619 / 0.136341
Separate direct ridge	0.080007 / 0.092130	0.069333 / 0.077874
Training mean	0.124287 / 0.137876	0.140423 / 0.143218

Lower RMSE is better. These are two differently normalized panels, so absolute errors are not comparable across columns. The primary full-ridge recipe was chosen on development, not selected retrospectively from protected results. Direct ridge has lower protected pooled error on Wessels and is reported as a predeclared additional control; it does not replace the primary comparator after observing outcomes. The original unrepaired Norman candidate has pooled RMSE 0.149550 and fails functional exchange invariance with maximum swap difference 5.794007 on protected inputs. The constructed repair has zero swap difference, but remains predictively inferior to the complete conventional alternative.

The prespecified criterion fails in both families. In Norman, the candidate's median paired relative gain against additive is 5.496%, but pooled error is worse and its ninetieth-percentile relative harm is 20.528%, exceeding 10%. Against full ridge its median paired gain is -8.003% and ninetieth-percentile harm is 49.754%.

These paired ratios are not the ratio of the tabulated marginal medians. Against the ordinary mean-residual control, median gain is only 0.140% and pooled error worsens.

In Wessels, the inherited construction passes the isolated additive contrast with median paired gain 11.716% and ninetieth-percentile harm 1.265%. It fails against MatchingMean, with median gain -10.574%, and against the primary full ridge, with median gain -77.941% and ninetieth-percentile harm 116.280%. Against the residual-mean control its median gain is 1.865%, below the required 5%. Thus a comparison restricted to additive would yield a misleading positive conclusion.

The Wessels old/new policy arms with the new toolbox, ordinary equivalent reuse and history-best action ordering have identical prediction hashes and outcomes. Operator removal, frozen/reset toolboxes and both old-tool policy arms produce the same additive fallback. The acquired operator is exercised on fresh work, yet its extra acquisition and verification costs do not purchase superiority over equivalent ordinary reuse or full ridge. This experiment does not establish either combination advantage or progressive lifecycle advantage.

The secondary observed-deviation ranking agrees with the boundary: candidate mean NDCG@50 is 0.3624 in Norman and 0.4042 in Wessels, versus MatchingMean 0.4783 and 0.5394, and full ridge 0.4229 and 0.7741. All pairs have defined rankings. These are finite-panel ranking measurements, not biological interaction-significance tests. Leave-one-pair and leave-one-constituent sensitivity summaries remain in the raw reports; they do not eliminate shared-data dependence or create a population-level confidence claim.

The evidence-directed decision is to retain the qualified conventional method for this delivery and the acquired repair for its functional scope. The current representation does not justify further favorable-result search on these exposed panels. A subsequent scientific experiment must state a new unresolved capability and a new independently justified test.

5. Reproducibility defects are evidence

This engineering repair edition preserves the earlier public paper, whose PDF SHA-256 is c27384332c 59b02ef3f27f450cbe81696de91d07a443c366827240dd2642f0bd, and its original rendered pages. The protected experiments, candidate freeze and all nine decisions remain unchanged. The revisions below concern operational enforcement rather than a new scientific attempt.

Ordinary unzipping of the preceding release retains `ARCHIVE_MANIFEST.json`. The self-packaging test then refused `ARCHIVE_RESERVED_MANIFEST`, producing 93/94 rather than the 94/94 obtained from the validated extractor, which consumes packaging metadata. The original archive and failure are preserved. The supporting repair validates the retained manifest, exact payload set and payload hashes before excluding only verified packaging metadata from repackaging. Added, missing or corrupted payloads continue to refuse. This does not relax the historical source identities.

The PR #9 CI run failed its interruption harness even though the displayed result reported no surviving children. A separate deterministic regression reproduces a valid interruption with zero adopted descendants. Other harness attempts could observe a worker too late to establish a live-worker precondition. Invalid harness attempts are preserved rather than called candidate failures. The revised check establishes a live-worker barrier and stable process identity before interruption, then requires that exact worker to exit, no remaining children after reaping any adopted descendants, an interruption marker, no false completion and unchanged replay refusal. Python can legitimately reap its own interrupted subprocess before the outer verifier adopts it; a positive adoption count is therefore not itself an invariant. The new deterministic regression establishes a possible false-negative schedule, not the unrecovered exact scheduling cause of the original CI run. Both the old failure and invalid harness attempts remain available. A benign namespace probe showed why host `/proc` identifiers and namespace-local syscall identifiers must be mapped explicitly when task-children files are unavailable. This is Linux qualification, not host-crash durability.

Review of the new connected entry point also found an initialization race after exclusive directory creation but before the process recorded ownership. A real interrupt could leave an owned directory without its interruption marker. The repair blocks SIGINT/SIGTERM only across that ownership-critical section, restores the prior mask and records the resulting pending signal as incomplete. Five regression tests include

actual pending-SIGINT delivery. The old executing source, deterministic defect and repair record are preserved.

The first full offline replay rebuilt the historical baseline, generated construction, mapping, repair, acquired memory, fresh-family search and all controls, then correctly refused when the independent judge found an edited frozen documentation file. Source identity was restored from the original registered bytes; the changed documentation became a separate edition. This failure is retained and is not a new scientific attempt or a reason to relax identity checks.

The second bounded replay completes the connected executing core: exact predecessor, generated construction, mapped criticism-to-repair, source-bound acquisition and retrieval, all nine fresh Wessels construction arms, every frozen prediction/control, exposed evaluation and both same-report settlement qualifications. It reproduces the scientific failures rather than turning replay into another confirmation. Its enclosing receipt records 128.457 wall seconds, 127.526 waited-child CPU-seconds and 64.059 supervisor CPU-seconds, with no surviving descendants. During that run an unused archive checker/test changed after the startup source check; the drift is separately recorded while the 761-file candidate freeze remains unchanged. This is an executing-core replay, not a claim that the final publication archive has already passed clean extraction and release qualification. The final archive requires its own source-matched checks.

Public review subsequently challenged two additional supervisor behaviors. Sending SIGTERM to the original standalone resource-wrapper process could leave its child running without a final receipt. A child that rapidly exceeded the output-storage allowance and exited between polling steps could receive a complete result. These are consequential operational defects; ordinary successful runs did not disprove them. The original wrapper remains byte-identical because it is part of the protected scientific source closure. The active CLI instead names a separately versioned replacement, preserving both the old source and the original CLI. The unchanged historical runs are not relabeled as having used the repair.

An independent namespace-aware replay establishes a live sleeping worker after the old supervisor exits with -15 and leaves no receipt; a separate rapid-exit run records complete with 15,676 output bytes against a 10,000-byte allowance. The initial inconclusive PID observation and another run that correctly caught an overrun are retained alongside these decisive counterexamples. A blocked accounting ledger does not make a falsely complete run acceptable.

The version-3 supervisor installs explicit termination handling, protects initialization and settlement critical sections, terminates and reaps the owned worker group, and checks final aggregate output including the receipt before classifying completion. Four wrapper regressions and five CLI regressions pass, including actual pending SIGTERM, interruption of a live worker and rapid-exit storage overflow. Independent source review confirms that the repaired version preserves the original frozen wrapper rather than rewriting its evidence. Its source hash and the active CLI are separately bound in this edition. The active CLI uses a 240-CPU-second outer reservation within the original finite campaign budget. These targeted tests qualify the repaired cases; they do not claim a new complete connected replay or final archive qualification before that later execution occurs.

6. Commerce and the remaining economic boundary

Public service schedules can constrain a conditional price model, but an unrelated analysis package is not a buyer contract for this instrument. No independently admitted customer cohort or willingness to pay is inferred from predicted laboratory savings. Existing protected exchange code is reused; a new ledger is not substituted for it.

The executed commerce adapter accepts the actual Wessels audit report only after matching a prefrozen functional contract, independent verifier source, obligations and artifact receipt. Acceptance concerns truthful source-bound reporting of every control, failure and null; it does not require a scientific win and is not buyer acceptance. The same bytes traverse the existing protected-exchange interface and payment-gated encrypted release. All 15 checks pass, including positive clearance, exact artifact download, duplicate settlement with no additional change, forged-binding refusal, nonpayment, acceptance disputes, interruption rollback/resume, correlated reversals and a separate liquidity falsifier.

The scenario price is an arbitrary 2,000,000 simulated microUSD, with eight arbitrary 10,000-unit cost categories per admitted case. They are functional stress inputs, not inferred market prices or complete measured costs. The favorable seven-case scenario has five releases, collections 10,000,000, scenario expenses 560,000 and scenario profit 9,440,000. Its opening capital is 20,000,000; cash is 29,440,000; tail obligations and working capital are separately represented. The ledger's unrestricted-earned measure is 9,440,000, while its stricter earned-control allocation remains zero. These different controls are not silently reconciled by treating capital as earnings.

After the correlated reversals, that same fixture has zero collections/earnings and loss 560,000; investment stops. The separate liquidity cohort has positive accounting profit 1,900,000 but zero available earned cash and refuses investment. Cohorts and favorable checkpoints are never pooled into a commercial result. The commerce integration took 3.830 wall seconds and 3.029 waited-child CPU-seconds; full monetary cost remains unknown.

No independent buyer cohort is admitted. Artificial fixture balances therefore establish zero commercially qualified earned funds and no funded successor. The Wessels technical successor was constructed before this money simulation and cannot be relabeled as earned-funded construction. Demand, the offering's price, conventional complete-service profit and the unknown historical cost term remain unqualified.

Measured execution resources do not provide full monetary cost. Agent inference, acquisition, integration, maintenance, support and remaining obligations are not zero merely because their bills are unavailable. Conditional break-even frontiers can expose an unknown-cost term, but cannot certify positive complete-cost profit. Refundable deposits, reserve releases and borrowing do not finance a claimed earned-funded successor.

7. Evidence interpretation

Engineering qualification, a consequential enabling capability, combination advantage, experience advantage and economic progression remain separate decisions.

Decision	Evidence-based verdict
Engineering qualification	Demonstrated in the declared finite construction, repair, retention and fresh-evaluation scope
Consequential enabling capability	Not established; strong conventional prediction remains available and superior
Combination advantage	Not established against the complete qualified controls
Experience advantage	Not established over equivalent ordinary reuse after acquisition costs
Conditional modeled commercial viability	Unknown without an admitted cohort, price and bounded full costs
Investment attractiveness	Unknown; earned-funded investment deferred
AGI ALPHA value capture	Unknown; beneficiary value is not an established receipt
Earned-funded successor benefit	Not established; no commercially admitted earned funds
Live commercial success	Not established; no actual orders or transfers

Functional symmetry repair and qualified data access support specific executable constructions. Neither supplies the missing scientific or economic links. The scoped engineering verdict does not retroactively certify every historical supervisor behavior or waive final source-matched release gates.

The current experimental restriction is a finite supplied language with developer-authored primitives and current-role outcome protection. Shared host, model family, solvers and libraries remain dependencies. Separate builder and verifier responsibilities constitute procedural separation, not independent external

certification. The paper records executed evidence and unresolved boundaries; it does not treat the mission's ambition as a demonstrated property.

The next decisive commercial step is an independently admitted cohort with a qualified useful advantage and bounded full costs. A settlement fixture cannot supply that evidence. New physical or biological instrumentation would also require operational validation beyond this condition-mean pilot.

8. Executing interfaces and evidence scope

Requirement	Executing interface	Test or receipt	Limit
Structural transfer drives a transformation	`exchange_mapping`; `repair`	Mapping and constraint records; independent symmetry adjudications	Supplied group-averaging primitive and role schema
Bounded generated strategies	Policy interpreter and broker	Generated policy trace and actual four-cell search	Finite arithmetic grammar; selected policy ties incumbent
Acquired generative inheritance	`acquire`; retrieval	Recomputed pre/post defect; immutable capability; Wessels refits	Ordinary reuse receives equivalent content
Protected scientific judgment	Frozen checker	Frozen predictions; independently reproduced controls; pair-level outcomes	Fixed-cohort descriptive pilot; unavailable complete models remain limits
Exact historical reproduction	`materialize`; archive	Actual bundled-release hash, 50 old sources, ordinary-extraction regression	Old freezes remain unchanged
Conditional delivery and finance	Commerce adapter	Independent artifact-acceptance binding and protected-exchange scenarios	No admitted commercial cohort or established complete-cost surplus

The `inheritance.py` module supplies the constraint and retrieval interfaces. The exact source paths, source hashes and function identities are retained in the mapped repair, acquired-capability, policy-cross and final-verifier receipts. The bounded entry point checks this source closure before execution.

On a qualified Linux host with the pinned Python dependencies and seccomp support, the bounded offline entry point is:

```
python -m pip install -r requirements-cycle004.txt
python run_cycle006.py --output /tmp/agi-alpha-cycle006-new
```

The output path must be new. The command checks the source closure before application imports, materializes the exact bundled predecessor, reconstructs mapping-driven repair, acquires and retrieves the operator, refits the actual fresh-family search arms, reproduces complete controls and evaluates the now-exposed frozen panels. It then executes same-report acceptance and conditional settlement. Historical native Omega proposals are explicitly replayed; new local generation is a separately labeled interface exercise. Reproduction is neither another protected confirmation nor evidence of live frontier-model discovery.

The implementation sources are readable at the public scientific revision. The edition's machine-readable input manifest binds the frozen experiment, raw receipts and reported results by exact hashes. The scientific revision contains the original protected results; the later connected replay and settlement receipts accompany this edition rather than being retroactively assigned to the earlier source checkpoint. Identity proves which

bytes were used; it does not certify their truth. The new source and evidence release must be evaluated separately from its predecessor.